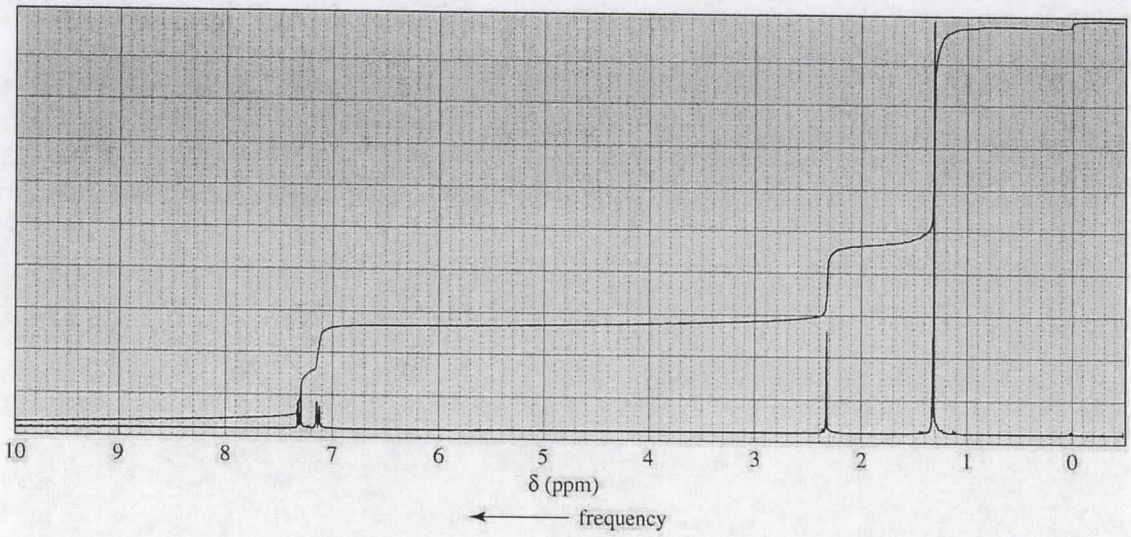
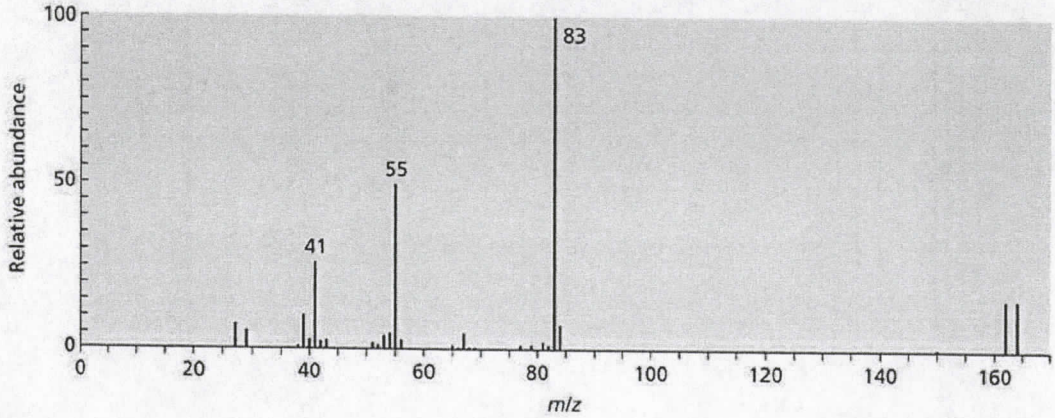
Organic Cumulative Examination

November 2015

1) Give the structure that corresponds to the following molecular formula and ^{1}H NMR data:

$C_7H_{16}O_4$: δ 1.93 (t, J = 6 Hz); 3.35 (s); 4.49 (t, J = 6 Hz); relative integral 1:6:1.
How would a ^{13}C NMR spectrum for this compound look?

2) Give a reasonable structure for a compound with the molar formula $C_{11}H_{16}$ and the ^{1}H NMR spectrum given here – explain how you derive your solution from the given information



3) Suggest a reasonable structure for a compound with the following MS spectrum – explain, how you derive the solution from the given information